

A low-temperature internal heating strategy without lifetime reduction for large-size automotive lithium-ion battery pack

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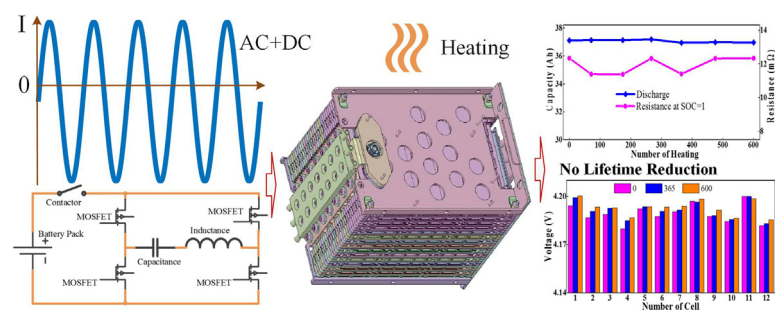
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HIGHLIGHTS

- A low temperature internal heating strategy with AC+DC for lithium-ion battery pack.
- The permissible AC and DC are determined to circumvent lithium-ion deposition.
- A simple soft-switching circuit with low loss is designed for heating battery pack.
- The essentially uniform temperature distribution within battery pack during heating.
- No lifetime reduction of battery pack after 600 repeated heating experiments.

GRAPHICAL ABSTRACT



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ABSTRACT

To overcome the long-standing challenge of poor performance of large-size automotive lithium-ion battery pack at low temperature, an internal self-heating strategy without lifetime reduction is proposed. A new method superimposing the discharge current on alternating current for self-heating is developed to prevent lithium-ion deposition, which not only avoids the measurement difficulty of potential and impedance of anode electrode, but also circumvents inconsistency problem of battery pack. The permissible alternating current and direct current are determined to avoid lithium-ion deposition. An effective yet simple soft-switching circuit is designed for heating of large-size automotive lithium-ion battery pack. The battery pack is warmed up from -20.8°C to 2.1°C within 600 s, where the temperature difference among twelve batteries is below 1.6°C , implying the essentially uniform temperature distribution. Based on the performance analysis for battery pack, no obvious loss of lithium inventory is found and no lifetime reduction is observed after 600 repeated heating tests due to the suitable over-potential of anode electrode and no substantial thermal-induced aging stress. The proposed self-heating strategy, validated for heating uniformly and effectively without lifetime reduction, is of high potential to deliver a practical solution to poor performance of large-size automotive lithium-ion battery pack in cold conditions.

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Nomenclature

C_k	thermal capacity ($\text{J}\cdot\text{K}^{-1}$)
C'_p	specific heat capacity ($\text{J}\cdot\text{g}^{-1}\cdot\text{K}^{-1}$)
Q_k	heat generation rate (W)
T_k	battery temperature ($^{\circ}\text{C}$)
R_x	thermal resistance ($\text{K}\cdot\text{W}^{-1}$)
R_k	thermal resistance ($\text{K}\cdot\text{W}^{-1}$)
T_a	ambient temperature ($^{\circ}\text{C}$)
I_{bk}	current through the battery (A)
U_{ocvk}	open circuit voltage (V)
I_{ack}	alternating current (A)
R_{ek}	the real part of impedance (Ω)
U_{dck}	direct voltage (V)
I_{dck}	direct current (A)
η_n	over-potential of anode electrode (V)
U_{nocv}	equilibrium potential of anode electrode (V)
i_F	faradiac current (A)
R_{ct}	charge-transfer resistance (Ω)
C_{dl}	constant phase angle element
W_D	warburg impedance (Ω)
R_b	battery ohmic resistance (Ω)
L_b	battery inductance (H)
U_{ocv}	open circuit voltage (V)
α_D	the differentiation order
α_d	the differentiation order
i_{dcF}	faradiac current (A)
i_{acF}	faradiac current (A)

R_o	system resistance of the soft-switching circuit (Ω)
L	inductance in the soft-switching circuit (H)
C	capacitance in the soft-switching circuit (F)
u_C	voltage of the capacitance (V)
U_{Co}	the initial voltage of the capacitance (V)
ω	the practical angle frequency (rad/s)
ω_o	the ideal angle frequency (rad/s)
δ	attenuation coefficient
β	attenuation angle (rad)
i_L	current through the inductance (A)
i_C	current through the capacitance (A)
T_{ac}	period time of alternating current (s)
f	the frequency of alternating current (Hz)
λ	weight coefficient
ΔT	temperature rise ($^{\circ}\text{C}$)
Q_{dis}	the discharge capacity (Ah)
i_{dis}	discharge current (A)

Abbreviations

AC	alternating current
DC	direct current
EIS	electrochemical impedance spectroscopy
EV	electric vehicle
IC	incremental capacity
LIB	lithium-ion battery
SOC	state of charge
SOH	state of health

1. Introduction

Although environmental friendliness, long lifetime, high energy density and high durability have made lithium-ion batteries (LIBs) as a dominant power source for electric vehicles (EVs), unsolved practical difficulties still remain [1,2]. Owing to low electrolyte conductivity, sluggish kinetics of charge transfer and decreased solid-state Li diffusivity for LIBs at low temperature, significant power loss and dramatically reduced driving range have proved especially troublesome for the expanding EVs application [3]. In addition, lithium-ion is inclined to deposit on the surface of anode electrode to form Li dendrite that may induce battery degradation and even safety hazards, if charging is forcibly conducted at extremely low temperature. As the kinetic and transport processes within LIBs depend highly upon temperature, battery performance recovers after being warmed up [4]. Heating LIBs is hence of vital importance for EVs operation in cold environment.

Compared with external heating, internal heating, which does not have the intermediate step of the heat propagation and ensures uniform temperature distribution, exhibits the advantage for fast heating LIBs with higher efficiency and less thermal-induced aging stress. There are plenty of previous attempts to investigate internal heating. Ji et al. [5] and Stuart et al. [6] found that alternating current (AC) heating is preferred because it heats the battery with little temperature difference inside the battery. But the systematical discussion on how to choose the heating parameters, including AC magnitude and AC frequency, to avoid battery degradation during the AC heating is not involved. Ge et al. [7] presented the temperature-adaptive AC heating method for lithium-ion deposition prevention. If the potential and impedance of anode electrode is known in advance and remains unchanged, the heating method is preferred. However, it is less possible to measure the potential and impedance of anode electrode, because the batteries in EVs usually do not include the third electrode. Furthermore, with the aging of LIBs, the potential and the impedance of anode electrode of LIBs inevitably change, resulting in the difficulty of applying this method. Ultimately, if there exists inconsistency in the battery pack, it

is a complicated task to select the heating parameters using this method. Thus, it is extremely critical to propose a new heating method to rapidly warm up battery pack without lithium-ion deposition.

Recently, a self-heating lithium-ion battery structure [8], where a nickel foil is embedded inside the LIB, is presented to enable rapid, efficient self-heating. Although the heating rate using this method is extremely high, there exists high temperature gradient within the battery, inevitably leading to thermal-induced aging stress and detrimental impact on battery health. A multi-sheet cell design is thus presented to achieve more uniform temperature distribution within the battery [9–11]. However, it comes with many challenges to manufacture this battery inserted with many nickel foils. Furthermore, it is difficult to implement this method for heating battery pack with multi-sheet nickel foil, owing to the tremendous difficulty of controlling the heating process. Most seriously, the voltage during heating is much lower than the normal voltage, inducing high direct current (DC) through the battery, which may result in dramatical damage on battery lifetime.

Previous attempts to investigate the heating method focus only on the small battery [4–10,12,13] or the individual battery [4,5,7–13]. The question on how to warm up the large-size automotive battery pack, which is of significant importance for EVs' applications, is still not well researched. Some heating methods are good enough for the individual battery or the small battery in laboratory but they are almost impractical for heating battery pack. It is a great challenge to apply the heating method with a third electrode [8–11] for heating battery pack due to the associated complication in manufacturing, as well as the inevitable damage effect on battery health. In order to quickly and safely heat the battery, a rapid low-temperature internal heating strategy with optimal frequency for LIBs [14] is proposed. Although the selection of adaptive AC magnitude [7] and optimal AC frequency [14] are discussed for the individual small battery, the question on how to select heating parameters for the large-size automotive battery pack remains unanswered due to the inconsistency problem in terms of the state of charge (SOC) and the state of health (SOH) among LIBs.

In this work, to circumvent inconsistency challenge of large-size

automotive battery pack, as well as to avoid the measurement difficulty of potential and impedance of anode electrode, a new method superimposing DC on AC for self-heating is proposed to prevent lithium-ion deposition. Based on the equivalent electrical circuit model, the current limitations of AC and DC are developed to avoid lithium-ion deposition during battery pack heating. For less loss and low complication, an effective yet simple soft-switching circuit is designed for heating of large-size automotive lithium-ion battery pack. An effective low-temperature internal self-heating strategy without lifetime reduction for battery pack is proposed and experimentally validated. The temperature distribution within the battery pack during internal heating remains essentially uniform with the temperature difference of less than 1.6 °C. The effect on SOH of the proposed heating method is experimentally verified using 600 repeated heating tests, and no lifetime reduction is observed.

2. Theoretical considerations

2.1. Thermal modeling for battery pack

A lumped thermal equivalent circuit model for battery pack as shown in Fig. 1 is developed to perform thermal analysis for fundamental insight into the internal heating process [15]. $R_a, R_b, \dots, R_l$ represent the thermal resistances between the battery and the surroundings, $R_1, R_2, R_3, \dots, R_{l1}$ are the thermal resistances between the neighboring batteries, and R_1, R_{l2} are the thermal resistances between connecting wires and the neighboring batteries. It is assumed that heat generation and heat dissipation mainly come from the core of the battery. $T_1, T_2, \dots, T_{l2}$ are the core temperatures of different batteries and T_0, T_{l3} describe the temperatures of connecting wires. The different temperatures are represented by potential at different locations and the heat generation rate ($Q_1, Q_2, \dots, Q_{l2}$) is represented by the current source. Thermal capacity (C_k) of every individual battery, used to describe the storage of heat energy in the battery, can be expressed by a capacitor. The core temperature of every battery can be expressed as

$$C_k \frac{dT_k}{dt} = Q_k + \frac{T_a - T_k}{R_x} + \frac{T_{k+1} - T_k}{R_k} + \frac{T_{k-1} - T_k}{R_{k-1}}, \quad k = 1, 2, \dots, l; x = a, b, \dots, l \quad (1)$$

Because the heat generation and the temperature of every battery are almost the same, the heat transfer process between the neighboring batteries can be ignored. Generally, the temperature difference between the connecting wire and the neighboring battery is low. For simplicity, it is assumed that there is no heat transfer between the connecting wire and the neighboring battery. Thus, the last two items on the right side of Eq. (1) can be neglected when calculating the battery temperature. The heat generated from the electrical operation of LIBs includes both entropic heat and electrical loss. However, the reversible heat is much smaller than the irreversible heat due to the high current during internal heating [16] and high resistance at low temperature. Thus, for simplicity, the reversible heat is neglected and the heat generation rate is deduced as

$$Q_k = \frac{1}{2} I_{ack}^2 R_{ek} + [U_{ack} - U_{ocvk}] I_{ack} \quad (2)$$

The irreversible heat is composed of the heat from AC and DC when the battery pack is heated. The heat generation rate depends on the AC frequency, the AC magnitude and the DC magnitude. The heating parameters, like the AC parameters and the DC magnitude, should be carefully selected to ensure a high heat generation and to avoid lifetime damage.

2.2. The analysis on lithium-ion deposition prevention

In order to avoid lifetime reduction of battery pack during internal heating, the difference between the solid phase potential and the

electrolyte phase potential of anode electrode should be higher than the reaction potential of lithium-ion deposition, which is 0 V for $\text{Li}/\text{Li} + [\text{7,17,18}]$. Thus, to prevent lithium-ion deposition on the anode electrode, the over-potential of anode electrode during charging is required as

$$|\eta_n| < U_{nocv} \quad (3)$$

Linearizing the Butler-Volmer equation [19,20], the over-potential approximately equals to

$$|\eta_n| \approx i_F R_{ct} \quad (4)$$

Therefore, the fundamental factor that determines the lithium-ion deposition during charging is the magnitude of the faradaic current, which is the current generated from the charge-transfer process. Based on the equivalent electrical circuit model in Fig. 2, the faradaic current can be calculated as

$$i_F = \text{abs} \left(\frac{W_D(j\omega)^{\alpha_D}}{R_{ct} C_{dl} W_D(j\omega)^{\alpha_D + \alpha_d} + W_D(j\omega)^{\alpha_D} + C_{dl}(j\omega)^{\alpha_d}} \right) i_b \quad (5)$$

When the AC frequency is high, the impedance of C_{dl} is low. If the frequency is high enough, the impedance of C_{dl} is extremely low and approximates zero, resulting the all current through the path of C_{dl} . Therefore, the higher the AC frequency is, the lower the faradaic current is. The open circuit voltage of the anode electrode depends on SOC. Thus, to avoid lithium-ion deposition at any SOC, the AC frequency should be selected to be extremely high. At high frequency, only the ohmic resistance generates heat and the real part of impedance is the least. Under the constant AC magnitude, the heat generation rate is the least at high frequency. Therefore, it is critical to take new method to use the high value of the large real part of impedance at low frequency to rapidly heat the battery without lifetime reduction.

If DC for discharging is superimposed, the faradaic current for charging can be significantly decreased. The over-potential of anode electrode in the charge-transfer process with AC and DC during charging can be expressed as

$$\eta_n = (i_{acF} - i_{dcF}) R_{ct} \quad (6)$$

As long as the anode potential is above 0 V, there is no lithium-ion deposition at any SOC [7,17,18]. Under a stable state, the DC current goes through the charge-transfer resistance and no current goes through the capacitance. Thus, the over-potential of the anode electrode in the charge-transfer process during charging can be deduced as

$$\eta_n = \text{abs} \left(\frac{W_D(j\omega)^{\alpha_D}}{R_{ct} C_{dl} W_D(j\omega)^{\alpha_D + \alpha_d} + W_D(j\omega)^{\alpha_D} + C_{dl}(j\omega)^{\alpha_d}} \right) i_{ac} R_{ct} - i_{dc} R_{ct} \quad (7)$$

Lithium-ion deposition during internal heating can be circumvented at any SOC under different frequencies, as long as Eq. (8) is satisfied.

$$\frac{i_{dc}}{i_{ac}} \geq \text{abs} \left(\frac{W_D(j\omega)^{\alpha_D}}{R_{ct} C_{dl} W_D(j\omega)^{\alpha_D + \alpha_d} + W_D(j\omega)^{\alpha_D} + C_{dl}(j\omega)^{\alpha_d}} \right) \quad (8)$$

The new heating method with AC + DC has several salient features. First, there is no need to consider the potential and impedance of anode

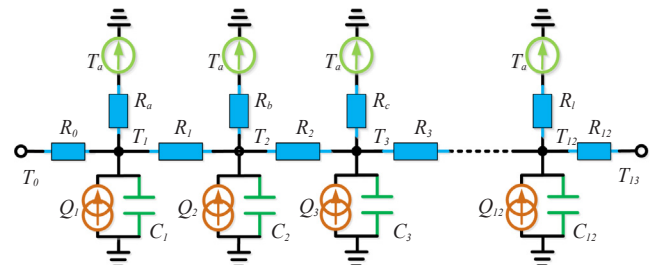


Fig. 1. A lumped thermal equivalent circuit model for battery pack.

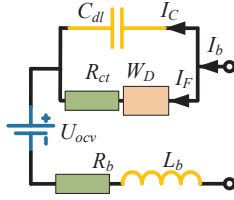


Fig. 2. Equivalent electrical circuit model for LIBs.

electrode. Although SOC of LIBs within the battery pack are different, the heating method is effective to prevent lithium-ion deposition. Second, there exists safety margin to avoid lithium-ion deposition when calculating current using Eq. (8) with the impedance of the full battery instead of the impedance of anode electrode. Actually, determination with the impedance of the full battery effectively circumvents the difficulty of impedance measurement of anode electrode. Ultimately, the high resistance at low frequency contributes to increase the heat generation rate for short heating time when superimposing DC for discharging.

2.3. Soft-switching circuit for heating

Compared with other circuit for generating high AC, the soft-switching resonance circuit as shown in Fig. 3 exhibits several advantages [21,22]. First, the current approaches zero when the switches are turning on or off. Second, the switching frequency is low and requirement for switching device is not high. Third, the loss of the switching device is low and heat generated from the switching process is almost zero. Fourth, the additional electrical source is not needed and the circuit can work by itself. Last but not least, the AC magnitude increases when battery temperature rises, which helps to rapidly increase the temperature of battery pack. Therefore, the soft-switching resonance circuit is selected to internally heat the large-size automotive lithium-ion battery pack.

When the total resistance of the circuit satisfies $R_o < 2\sqrt{L/C}$, the resonance occurs between the inductance and capacitance. Because of the existence of the resistance, the current magnitude of the oscillation exponentially damps. If no measure is taken, the current gradually turns to zero. So the full bridge switching-mode for the resonance circuit is developed. When an oscillation period is achieved, a pair of switches turn off and another pair of switches turn on. The new oscillation period occurs. When the oscillation period is completed, a pair of switches turn off and another pair of switches turn on. The continuous current without great damping is obtained and the current through the battery pack approximates the sinusoidal current. The voltage of capacitance can be expressed as

$$u_C = \frac{(U_{C0} - U_{ocv})\omega_0 e^{-\delta t}}{\omega} \sin(\omega t + \beta) + U_{ocv} \quad (9)$$

where

$$\delta = R_o/(2L), \quad \omega = \sqrt{1/(LC) - (\delta)^2}, \quad \omega_0 = \sqrt{\delta^2 + \omega^2}, \quad \beta = \arctan(\omega/\delta).$$

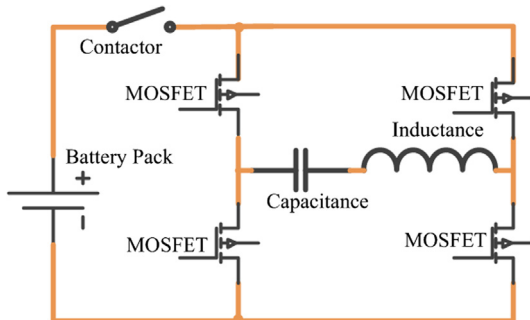


Fig. 3. A soft-switching resonance circuit.

Define $\varphi = (U_{C0} - U_{ocv}) \cdot \sqrt{C/L}$. The current through the inductance and capacitance can be deduced as

$$i_L = i_C = \varphi e^{-\delta t} \sin(\omega t) \quad (10)$$

The current through the battery pack can be expressed as

$$i_b = \begin{cases} i_L & 0 \leq t < T_{ac}/2 \\ -i_L & T_{ac}/2 \leq t < T_{ac} \end{cases} \quad (11)$$

The higher the open circuit voltage of battery pack and the capacitance value are, the higher the peak value of the current through battery pack is. The current through battery pack increases with the decrease of the inductance value. The peak and valley values of current of battery pack are $i_{b,p} = \varphi \exp[-\delta/(4f)]$ and $i_{b,v} = -\varphi \exp[-3\delta/(4f)]$, respectively. The heating current not only includes AC component, but also contains the DC component. When $R_o \rightarrow 0$, the DC approaches zero. The DC magnitude can be approximately calculated as

$$i_{dc} = (i_{b,p} + i_{b,v})/2 \quad (12)$$

The AC magnitude can be approximately expressed as

$$i_{ac} = (i_{b,p} - i_{b,v})/2 \quad (13)$$

2.4. Internal heating strategy for battery pack

To rapidly heat the battery, the high AC magnitude are required because of no substantial change of SOC. So the high DC is needed to circumvent lithium-ion deposition. However, the high DC is not desired due to the increased discharge capacity. Therefore, a good tradeoff between rapid heating (high temperature rise) and less loss of discharge capacity needs to be carefully determined, as shown Eq. (14).

$$y = \lambda \Delta T + (1-\lambda)Q_{dis} \quad (14)$$

$$\text{with } \begin{cases} \frac{dT_k}{dt} = \frac{Q_k}{C_k} - \frac{T_k - T_a}{R_k C_k} \\ \Delta T = \sum \frac{dT_k}{dt} t \\ Q_{dis} = \sum i_{dc} t \end{cases}$$

The temperature-rise rate (dT_k/dt) is selected as the indicator for heating battery pack. When temperature-rise rate and loss of discharge capacity is pre-set, the heating parameters can be determined. The low-temperature internal heating strategy for battery pack using AC + DC is proposed to quickly warm battery pack up without lithium-ion deposition.

3. Experimental

The test sample is a commercial laminated battery (capacity 35 Ah, weight 1010 g, cut-off voltages 4.2/3 V), consisting of graphite anode material and $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ cathode material. The large-size automotive lithium-ion battery pack (capacity 35 Ah, cut-off voltages 50.4/36 V) is composed of twelve individual batteries, where they are connected in series. The experimental platform, as shown in Fig. 4, includes the testing system and the monitoring system. The testing system consists of battery test equipment (Arbin BT2000) for testing the performance of battery pack, a climate chamber providing controllable temperature, and the soft-switching resonance circuit designed to heat the battery pack. The monitoring system, monitoring and logging voltage, current and temperature data, contains a current probe, a digital oscilloscope and a data acquisition unit. The eight T-type thermocouples are affixed in center parts of the chosen eight batteries (Number of the batteries: #1, #2, #4, #6, #7, #9, #11, #12) within the battery pack. The thermocouples has been calibrated to achieve 0.1 °C precision before experiments.

Four individual batteries with different capacity (SOH) were adjusted to 0.5 SOC for the measurement of electrochemical impedance spectroscopy (EIS) using the electrochemical workstation under



Fig. 4. Experimental platform for heating and testing battery pack.

different temperatures, as shown in [Table 1](#). The batteries were placed at each temperature in the climate chamber for more than 10 h for thermal equilibration. Other EIS measurements were conducted over a SOC range from 0.2 to 1 with intervals of 0.2 at -20°C and 4°C . EIS experimental conditions were: perturbation current, $I_a = 1\text{ A}$ over the frequency range from 10 mHz to 10 kHz, for twenty points per decade.

Before internal heating, the battery pack was soaked at -20°C in the climate chamber for more than 10 h for thermal equilibration. The battery pack was preheated from -20°C using AC + DC generated from the designed soft-switching resonance circuit. The heating process ends when average temperature of LIBs reaches above 0°C or after 10 min of heating. In order to investigate the influence of the internal heating on battery health, the battery pack is repeatedly heated with AC + DC for 600 times. Before the repeated heating, the tested battery is soaked at 25°C for 10 h and then the performance test is carried out as shown in [Table 1](#). The heating tests are performed for 10 min and then the battery pack rests for more than 3 h. If the voltage of each battery is higher than 3.45 V during internal heating, the heating process continues as shown in [Fig. 5](#). The charging test is carried out as shown in [Table 1](#), when the voltage of certain battery is lower than 3.45 V during internal heating. After the battery pack has been heated for about 70 times, 100 times or 125 times, the tested battery pack is soaked at 25°C for 10 h, and then the performance test was performed.

The other experiments are performed to obtain the specific heat capacity (C_p) of the individual battery and the thermal resistances between the different individual batteries and the surroundings in the lumped thermal model for battery pack. Two individual batteries are warmed up under adiabatic condition using accelerating rate

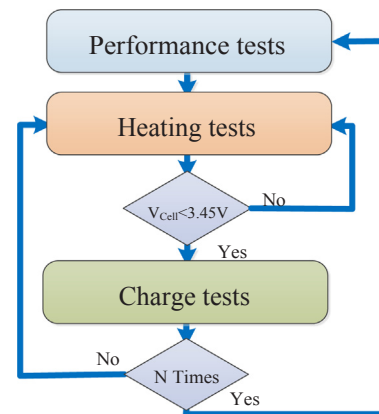


Fig. 5. Flowchart of lifetime test experiment for heating battery pack.

calorimeter, to calibrate the specific heat capacity. The recovery of battery temperature after heating test at -20°C is adopted to estimate thermal resistances.

4. Results and discussions

4.1. Suitable heating parameters

In order to avoid lithium-ion deposition during internal heating, the ratio between DC and AC magnitude is experimentally determined. Fig. 6 shows the evolution of the ratio between DC and AC magnitude

Table 1
Experimental details.

Experimental details.			
	Test	Item	Value
Battery pack with twelve batteries	Heating tests	Temperature	− 20 °C
	Charge tests	Charge, 25 °C	1/2C charge until 49.5 V; 1/4C charge until 49.5 V
	Performance tests	Charge, 25 °C	1/3C charge until 4.2 V; 1/6C, 1/12C, 1/18C, 1/30C charge until 4.2 V
		Discharge, 25 °C	1/3C discharge until 3 V
		Charge, 25 °C	1/10C charge until 4.2 V
Four individual batteries	EIS tests	Temperature	− 20 °C, − 8 °C, 4 °C
		Different SOC	1, 0.8, 0.6, 0.4, 0.2, 0.1; at − 20 °C, 4 °C

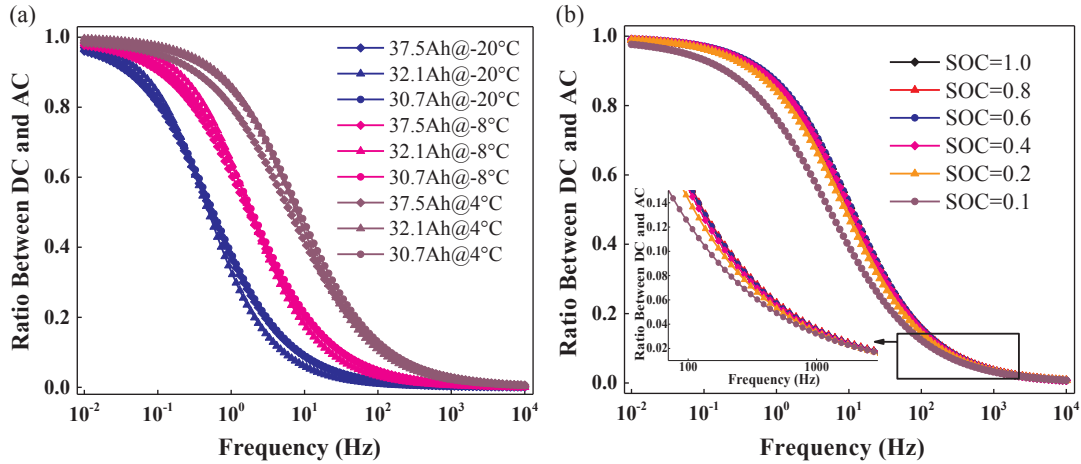


Fig. 6. The evolution of DC/AC ratio (a) under different SOH and temperatures (b) under different SOC at 4 °C.

under different SOH and SOC. The DC/AC ratio decreases with the increase of the AC frequency. At low AC frequency, the DC should almost approach the AC to prevent lithium-ion deposition. At high frequency, the DC can be close to zero. There is little change of the DC/AC ratio under different SOH, as well as under the SOC range between 0.2 and 1.0. The DC/AC ratio increases, especially at the middle frequency, when battery temperature rises. At low SOC, the DC/AC ratio decreases. The evolution of the DC/AC ratio under different SOC, SOH and temperatures indicates that DC should be low at high frequency and extremely high at low frequency to circumvent the reduction of battery lifetime.

Assuming the maximum current lower than 350 A, the heat generation rate with different DC magnitudes at different frequencies under -20°C and 4°C are displayed in Fig. 7(a). It is observed that the heat generation rate increases and then decreases with the increase of the AC frequency. This is because that the acceptable AC increases and then almost keeps at a constant value of about 350 A. At high frequency, despite almost the same AC, the resistance decreases, leading to the decrease of heat generation rate. There exists a peak of heat generation rate, which is helpful to reduce heating time. The peak value of heat generation rate increases and the frequency corresponding to the peak decreases with the increase of DC. The peak value of heat generation rate decreases and the frequency corresponding to the peak increases when battery temperature rises. If high DC is selected, the high heat generation rate and low AC frequency, as well as high loss of discharge capacity are achieved. Fig. 7(b) exhibits the evolution of the ratio of

heat generation rate between DC and AC with frequency under different DC magnitudes and different temperatures. Heat generation rate from DC dominates at low frequency while heat generation rate by AC accounts for the majority part at high frequency. If high loss of discharge capacity is not desired, heating with high frequency is ideal.

As for the individual battery, the heating parameters are easy to be selected. The heating parameters for battery pack, where the inconsistency always exists in terms of temperature, resistance, SOC and capacity among LIBs, should be carefully determined to achieve a good tradeoff between short heating time and less loss of discharge capacity. Based on the analysis of heat generation rate and discharge capacity loss, the loss of discharge capacity is lower than 7% and the temperature-rise rate is higher than $2^{\circ}\text{C}/\text{min}$, to optimally warm up the battery pack without lifetime reduction. Thus, the DC is about 14 A and the AC is about 336 A. Then, the parameters of the capacitance and inductance are determined as $1500\ \mu\text{F}$ and $29.76\ \mu\text{H}$, respectively. Accordingly, the AC frequency is determined as about 754 Hz. The AC and DC current flows through the twelve batteries, generating considerable heat, which rapidly warms up the series-connected large-size automotive lithium-ion battery pack without lifetime reduction.

4.2. Internal heating for battery pack

Based on the proposed heating method, the series-connected battery pack is heated at -20°C using the designed soft-switching resonance circuit, after placed at -20°C in the climate chamber for more than

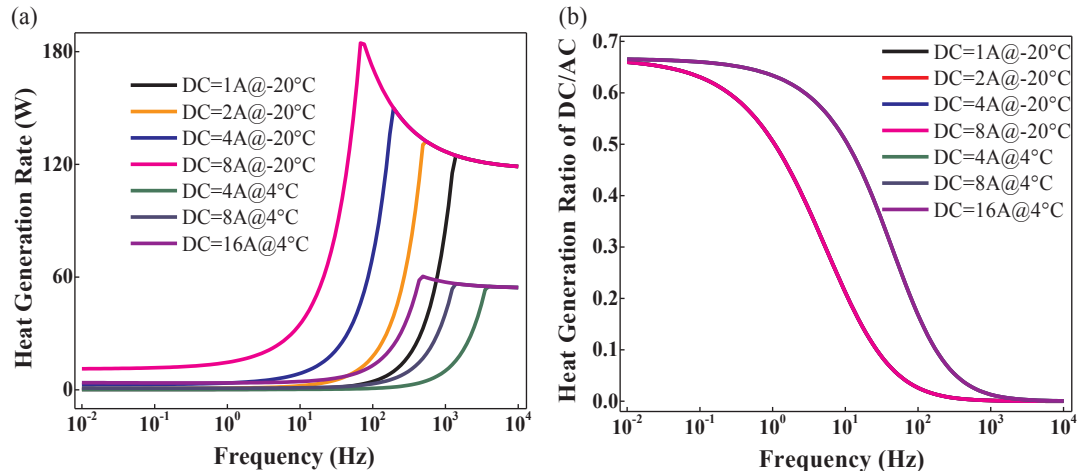


Fig. 7. (a) The evolution of heat generation rate with different DC under different frequencies and temperatures (b) The ratio evolution of heat generation rate between DC and AC at different frequencies.

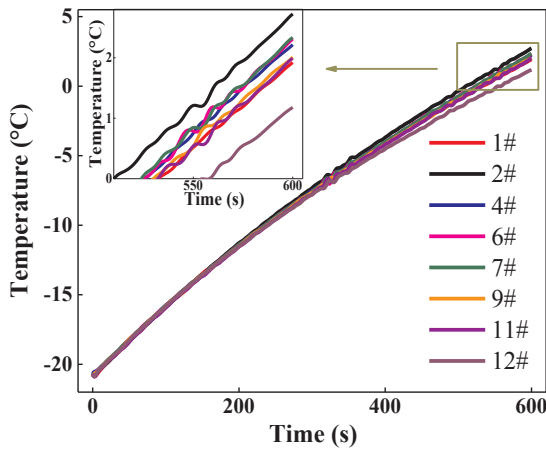


Fig. 8. The temperatures in the center part surface of eight batteries during internal heating.

10 h. The surface temperatures in the center parts of eight batteries during internal heating are shown in Fig. 8. It is observed that all of the temperatures increase consistently from -20°C during internal heating. The maximum temperature difference of below 1.6°C among the twelve batteries occurs in the end of heating, which mainly results from the high heat dissipation of the batteries at the outside. The small temperature difference of LIBs within the series-connected battery pack confirms that the proposed heating strategy can uniformly warm up the battery to a suitable temperature without high temperature gradient. The extremely low temperature gradient among the twelve batteries helps to avoid hot spots and keep good consistency of SOC among twelve batteries, thus prolonging the lifetime of series-connected battery pack.

Because of the low temperature difference, the mean value of eight batteries is taken as the temperature of the series-connected battery pack. Fig. 9 exhibits that the battery pack is heated from -20.8°C to

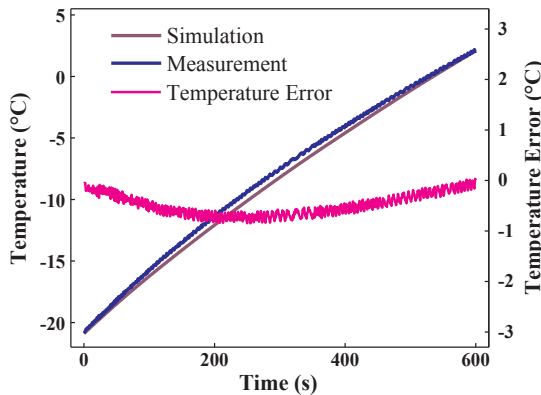


Fig. 9. Temperature evolution and error distribution during self-heating.

Table 2

The consistency analysis of the voltage and resistance of twelve individual batteries.

Number of heating		Mean	Variance
0	Voltage (V)	4.1892	0.0059
	Resistance ($\text{m}\Omega$)	0.939	0.059
365	Voltage (V)	4.1914	0.0054
	Resistance ($\text{m}\Omega$)	0.802	0.086
600	Voltage (V)	4.1929	0.0048
	Resistance ($\text{m}\Omega$)	0.903	0.062

Table 3

The measured specific heat capacity of the individual battery and thermal resistances between different individual batteries and surroundings.

Specific heat capacity ($\text{J}\cdot\text{g}^{-1}\cdot\text{K}^{-1}$)	Thermal resistance ($\text{K}\cdot\text{W}^{-1}$)							
	R_a	R_b	R_d	R_f	R_g	R_i	R_k	R_l
1.177	20.90	23.13	37.08	47.54	46.49	34.99	18.67	14.76

2.1°C within 600 s. The temperature of the battery pack as a function of time is almost a straight line up because of the growing AC magnitude, where the average temperature-rise rate is $2.29^{\circ}\text{C}/\text{min}$. Based on the identified parameters as shown in Table 3 in Appendix, the simulation result from the thermal equivalent circuit model is of great consistence with the experimental data, where the maximum absolute temperature error is below 0.9°C . The simulation error may come from the difference between the core temperature and the surface temperature or result from the neglecting of the reversible heat. The well-matched results indicate that the assumption for thermal model of battery pack is rational and the proposed heating strategy is effective.

Fig. 10 shows the evolution of voltage and current of battery pack at the beginning and the end of the heating process. The magnitude of AC current is low while the magnitude of AC voltage is high at the beginning of the internal heating process. The high resistance at -20°C leads to the high DC current of 19 A and low AC current of 330 A, which corresponds to Eqs. (12) and (13). On the other hand, the low resistance at 2°C leads to the high AC current and low DC current. The loss of discharge capacity is 6.64% during the initial heating at the SOC of 100%. The impact on driving range of EVs may induce a concern. Actually, in spite of the energy consumption during heating at -20°C , the driving range of EVs will increase by above 29% after internal heating (the related data is shown in Fig. 15 in Appendix). Despite the difference of AC and DC current, the limitations of DC and AC satisfy Eq. (8), implying no possibility of lithium-ion deposition during internal heating. The increasing AC current with the rise of battery temperature contributes to increase the heat generation rate, resulting in the short heating time.

The heating with 10 min for battery pack at different SOC as displayed in Fig. 11 indicates that the designed circuit can work well at different conditions. The temperature-rise rate decreases with the decrease of SOC, because of the decrease of the voltage and the current of battery pack. The temperature-rise rate changes from $2.29^{\circ}\text{C}/\text{min}$ to $1.86^{\circ}\text{C}/\text{min}$ as the SOC varies from 1.0 to 0.2.

4.3. The concern on battery pack health

Whether or not the high current at low temperature causes a damage on battery lifetime still remains a concern. Thus there are some

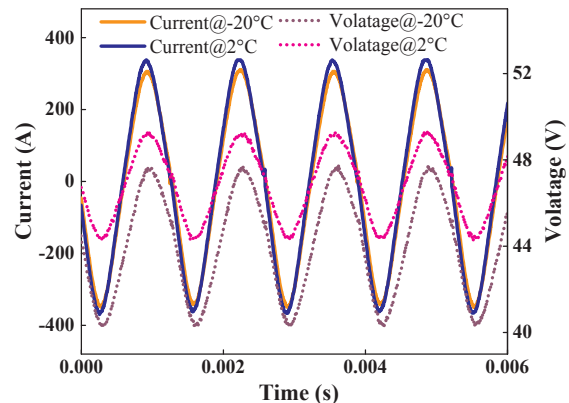


Fig. 10. Evolution of voltage and current of battery pack.

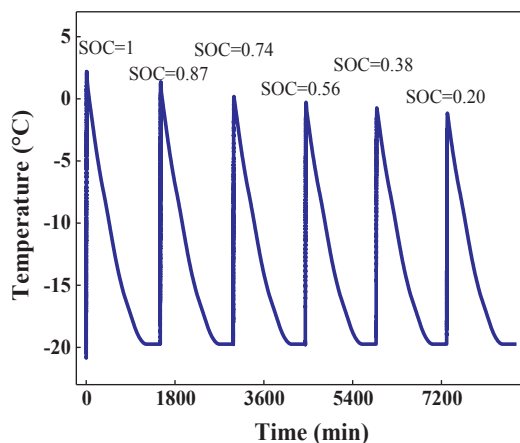


Fig. 11. The heating process at different SOC for battery pack.

performance tests, including capacity tests, resistance tests and charge tests with a small current, conducted to avoid such concern. The discharge capacity changes from 37.13 Ah to 36.97 Ah and decreases very little as shown in Fig. 12(a), after 600 repeated heating tests. The 1 s resistance at the SOC with value of 0 decreases gradually, which may be because that the AC with high frequency improves the active material utilization giving the battery pack lower charge-transfer resistance. The charge capacity also decreases very little as shown in Fig. 12(b) and the 1 s resistance at the SOC with value of 1 is kept almost the same during 600 repeated heating tests. As shown in Fig. 13, the incremental capacity (IC) curves are kept almost the same without obvious change in position and shape of any IC peak during 600 repeated heating tests, implying no obvious loss of lithium inventory [23–25]. This means that the proposed heating method can effectively prevent lithium-ion deposition. The capacity, resistance and IC curves vary very little after 600 repeated heating tests, indicating that the battery pack is not in accelerated deterioration. In other words, the proposed internal heating strategy has no detrimental effect on battery lifetime owing to the suitable over-potential and the faradic current of approximately zero.

Fig. 14 displays the voltage and resistance of twelve individual batteries at the end of the charge process under different numbers of internal heating. It is concluded that almost all of the voltages of the individual batteries increase while the resistances have no noticeable tendency of change. The resistances of some batteries, such as the battery #2, #7, #8, #9, #10, #11, decrease firstly and then increase with the increasing heating number. This may be because that the AC

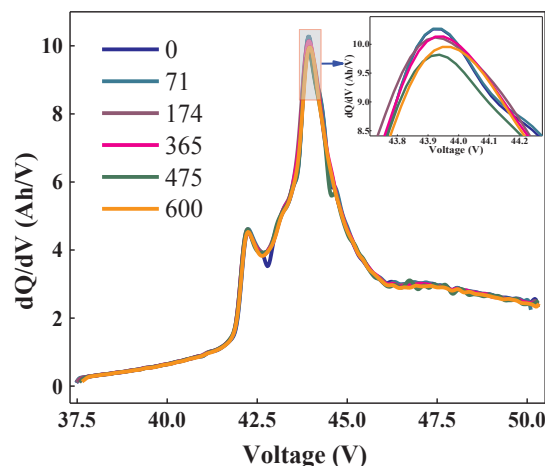


Fig. 13. The evolution of the IC curves during charging with 0.1C under different heating numbers.

with high frequency improves the active material utilization giving these batteries lower polarization. After internal heating, the average value of voltages increases while the variance value decreases as shown in Table 2. Good consistency of voltages of twelve individual batteries in the end of the charge process may result from good consistency of SOC of the individual battery due to the uniform temperature distribution within battery pack, besides the influence from the resistance of the individual battery. The consistency analysis indicates that the performance of the voltage and resistance of twelve individual batteries does not degenerate. This implies that the proposed heating strategy for series-connected battery pack has no detrimental effect on the SOH of LIBs.

To investigate the impact of the heating method on battery lifetime, instead of repeated heating for limited numbers [4,14,26], such as 30 times [14], 40 times [26], the battery pack has been repeatedly heated for 600 times. Since the influence of low-temperature heating on battery lifetime may be not obvious, the damage on battery lifetime using the limited heating numbers may not be observed. So, heating the battery for plenty of numbers should be considered and repeated heating for 500 times are conducted in the literature [8] to enlarge and accumulate the impact of low-temperature heating on battery lifetime. It is found that the capacity loss is about 7% after 500 numbers of repeated heating [8]. Nevertheless, only using the capacity to evaluate SOH of LIBs is less than objective [1], especially under high heating

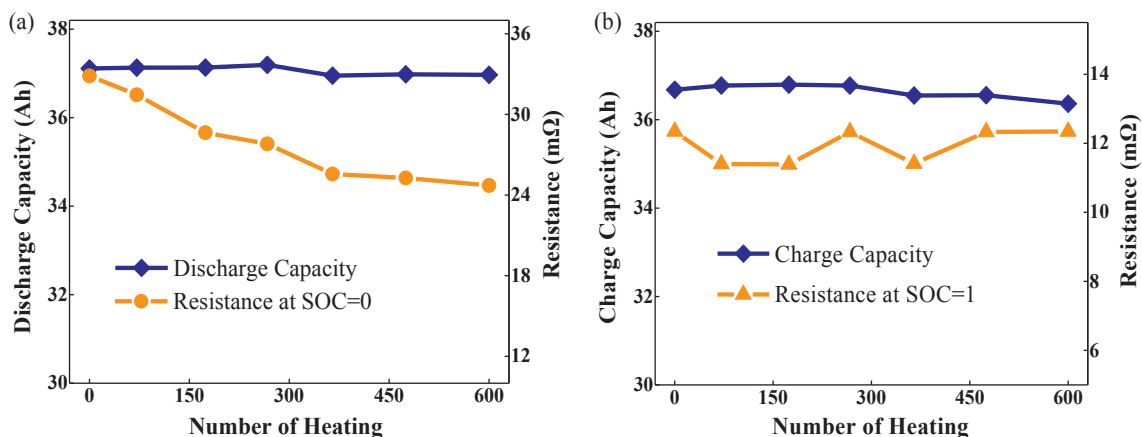


Fig. 12. The evolution of (a) the discharge capacity and resistance (b) the charge capacity and resistance.

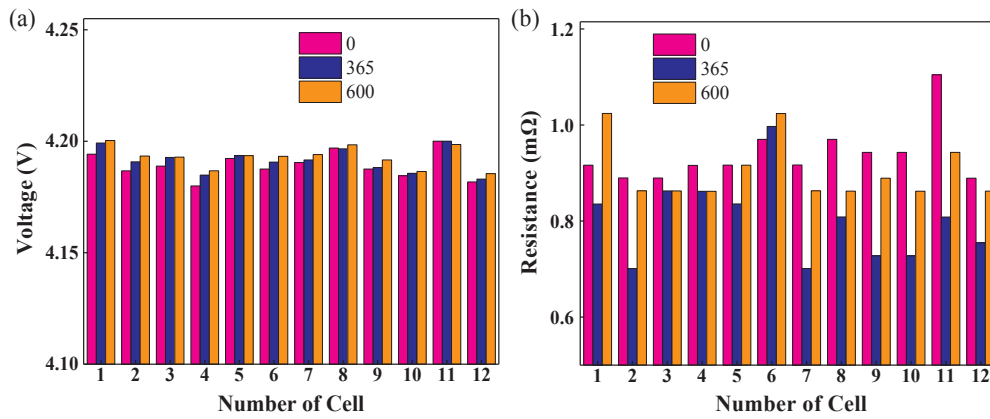


Fig. 14. (a) The voltage (b) Resistance of twelve individual batteries in the end of the charge process under different heating numbers.

current. It is rational to evaluate SOH of LIBs by investigating the variations of battery capacity and battery resistance, as well as the IC curves used to indicate the thermodynamics. Furthermore, for battery pack, the consistency of the voltage and the resistance of the individual battery is carefully compared and analyzed. The conducted experiments are sufficient to evaluate the influence of the proposed heating method on battery lifetime. The compared results demonstrate that the proposed heating method, using AC + DC to prevent lithium ion deposition, can heat battery pack to a suitable temperature without lifetime reduction at low temperature.

5. Conclusions

An effective low-temperature internal self-heating strategy without lifetime reduction for large-size automotive lithium-ion battery pack is proposed with the following salient features:

- Lithium-ion deposition prevention.** To prevent lithium-ion deposition, a new method with AC + DC for self-heating is proposed, which not only avoids the measurement difficulty of potential and impedance of anode electrode, but also circumvents inconsistency problem of battery pack. The equation of the current limitations, which is based on the equivalent electrical circuit model, is developed to avoid lithium-ion deposition for heating battery pack under different SOC and SOH, as well as different temperatures. Furthermore, with the addition of DC, the high resistance at low AC frequency can be adopted to heat battery pack with higher heat generation rate, contributing to shorter heating time.
- Heating uniformly and effectively.** Because of less loss and low complication, as well as no requirement of the additional electrical source, an effective yet simple soft-switching circuit is designed for heating of large-size automotive lithium-ion battery pack. The well matched experimental and simulation results confirm that the battery pack is heated from -20.8°C to 2.1°C within 600 s using the proposed heating method. The maximum temperature

difference is less than 1.6°C during internal heating, implying the proposed heating method can warm up battery pack to a suitable temperature with essentially uniform temperature distribution within the battery pack.

- No lifetime reduction.** Neither apparent variation of battery consistency nor noticeable capacity loss and resistance increase is observed after 600 repeated heating tests for battery pack. There is no obvious loss of lithium inventory from the nearly identical IC curves, implying that the proposed heating method can effectively prevent lithium-ion deposition. This is due to the suitable over-potential of the anode electrode, and the faradaic current for charging of approximately zero, as well as the essentially uniform temperature distribution within battery pack. The proposed heating strategy with suitably determined parameters is demonstrated to have no detrimental effect on the lifetime of large-size automotive lithium-ion battery pack, which yields great potential for improving operating performance of electrical vehicles in cold weather.

The repeated heating tests for 600 times indicate the good feasibility and high reliability of the designed heating system for battery pack. However, due to the limited space in hybrid and electric vehicles, the heating system installed beside the charging pile may be a good choice. Further study will be carried out on how to determine the magnitudes of the DC and AC so that the heat generation rate is higher while preventing the risk of lithium-ion deposition. The OCV characteristic at different temperatures will be considered when selecting heating parameters. Moreover, the comparison between the internal heating method and other heating method for battery pack is being studied.

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Appendix A. Appendix

Under different temperatures, lithiation and de-lithiation of LIBs are both carried out at the 1/3C rate with a 1 h rest in between, as shown in Fig. 15. The discharge capacities are 36.9 Ah, 29.5 Ah and 16.1 Ah at 25°C , 0°C , and -20°C , respectively. The discharge capacity at 0°C is 13.4 Ah more than that at -20°C . This implies that battery capacity will increase by 36.3% if the battery is heated to 0°C from -20°C .

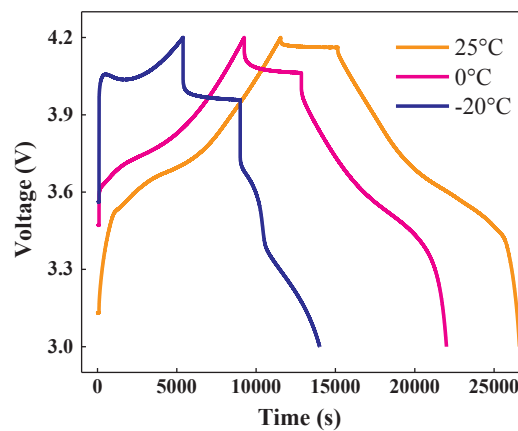


Fig. 15. The evolution of battery voltage during lithiation and de-lithiation with 1/3C at different temperatures.

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